PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Inflammation and nutrition-based biomarkers in the prognosis of esophageal cancer: A systematic review and meta-analysis
AUTHORS	Jiang, Yan; Xu, Dian; Song, Huan; Qiu, Beibei; Tian, Dan; Li, Zhongqi; Ji, Ye; Wang, Jianming

VERSION 1 – REVIEW

REVIEWER	McMillan, Donald University of Glasgow, Academic Unit of Surgery
REVIEW RETURNED	24-Feb-2021

GENERAL COMMENTS	A well presented comprehensive review of inflammatory and nutrition-related markers and their association with OS, DFS and CSS in esophageal cancer General comment The discussion would benefit from comment on how these results relate to the current GLIM criteria (Cederholm T et al) for diagnosing malnutrition in cancer.

REVIEWER	Darling , Gail E University of Toronto
REVIEW RETURNED	06-Mar-2021

GENERAL COMMENTS	1. As a consumer of this literature, it is helpful to have a comprehensive meta-analysis of these studies. It appears that some of the references are updates of previous reports. If so, generally only the most recent study is included. Please review references to ensure no duplication. 2.Do the authors have any recommendations for clinicians based on this study or was it primarily an academic exercise? How will clinicians use the study results? 3.Given some heterogeneity, it may be useful to include only sqaumous cell cancer patients in the analysis. 4.Although performance characterisitics of each test are reasonable, none appear robust enough to influence clinical decision making. Although not the intent of this study, did the authors consider combining tests to develop a more robust indicator?

REVIEWER	Escarela, Gabriel Universidad Autonoma Metropolitana, Departamento de Matemáticas
REVIEW RETURNED	03-Aug-2021

GENERAL COMMENTS	This systematic review and meta-analysis appears to have been well carried out and appropriately analysed. The results are presented in satisfactory detail and a suitably moderated interpretation of the data is made to take account of the acknowledged limitations of the study. The main limitation is the high degree of heterogeneity between studies, which could also be due to a number of factors which were not able to be assessed. In my (subjective) opinion the problems of bias could seriously affect the usefulness of this paper, but I acknowledge that if the subject matter is considered sufficiently important, then the paper could be published.

VERSION 1 – AUTHOR RESPONSE

#Reviewer 1:
Response:

We really appreciate your valuable suggestion. We have added the following sentences in the part of Discussion--"The Global Leadership Initiative on Malnutrition (GLIM) standards integrate current best evidence and expert opinion on malnutrition to promote the prevention, identification and treatment of malnutrition in cancer patients. Inflammation is one of the etiological criteria in GLIM classification, and studies have demonstrated that the changes of GPS score, C-reactive protein and albumin are highly consistent with the GLIM criteria in identifying malnutrition in patients. Similar to this study, our findings confirms the value of mGPS and CAR in predicting the prognosis of esophageal cancer. GLIM criteria are reevaluated every 3-5 years based on new research. Our results may provide a basis for the optimization of GLIM criteria. Additionally, previous studies have reported that the combination of prognostic nutrition index (PNI) and GLIM criteria has significant advantages in predicting the incidence and survival rate of perioperative malnutrition. More well-designed studies are needed to develop joint indicators to improve the sensitivity and specificity of prediction. "

#Reviewer 2:
Comments 1:
Response:
Thank you for your constructive suggestions. We carefully checked all the included literature.A total of 14 articles had common authors, which was suspected of duplication. After verification, we found that although the literature shared the authors they all targeted at different indicators without overlap, except for Feng JF (2014) and Feng JF (2019).The data of Feng JF (2019) and Feng JF (2014) are completely overlapping, and It is confirmed that Feng JF (2014) is the original text, so we removed the literature of Feng JF (2019) and modified all the parts involving this literature in the paper.

Sthdy	Sample size	Recruitment period	Indices
Hirahara N (2016)	147	2006.01--2015.02	LMR、NLR、PLR
Hirahara N (2018)	169	2006.01--2015.12	PNI
Feng JF (2014)	375	2006.01--2008.12	PNI

Feng JF (2014)	1048	2005.01--2008.12	GPS
Feng JF (2017)	298	2005.01--2008.12	SII
Feng JF (2019)	1048	2005.01--2008.12	GPS
Han LH (2015)	218	2007.01--2008.12	LMR、NLR、PLR
Han LH (2015)	206	2007.01--2008.12	PNI、SIS
Tian R (2016)	442	2005.01--2010.12	GPS
Tian R (2016)	260	2007.01--2011.12	CRP
Otawa Y (2016)	442	2007.01--2012.08	mGPS
Otawa Y (2017)	149	2007.02--2014.12	CRP
Lindenmann J (2014)	174	2003.01--2011.12	GPS
Lindenmann J (2017)	218	1999.01--2010.12	CRP

Comments 2:
Response:
Thank you for your constructive suggestions. We have added the following sentences in the part of Discussion--“ Our results of this meta-analysis will help clinicians and patients to select appropriate indicators for prognosis prediction, and in this way, patients can be classified and appropriate treatment strategies and postoperative management methods can be selected, which can provide ideas for policy makers. Taken mGPS for an example, esophageal cancer patients with a score of 2 may have a high risk of prognosis, which may provide an effective way for clinicians to select high-risk patients with worse prognosis or severe adverse events before treatment and further timely adjust individualized treatment regimens and enhance postoperative rehabilitation. ”

Comments 3:
Response:
Thank you for your constructive suggestions. We did subgroup analysis and meta regression, and found most of the sources of heterogeneity, such as sample size, gender distribution, truncation value, etc., but there were still some heterogeneity sources that could not be found.Tissue type may be the source of heterogeneity, but in this study, it was not found to be the source of heterogeneity according to the results of subgroup analysis and meta-regression. In addition, the purpose of this study was to analyze all tissue types, and the subgroup analysis would also show the combined effect value of each indicator in different tissue types, so we sincerely wish to report all tissue types.

Comments 4:
Response:
Thank you for your constructive suggestions. Accumulating literature have shown the prognostic values of inflammation and nutrition-based biomarkers, but with inconsistent findings. Therefore, we conducted this study to evaluate and compare the role of various indicators in prognostic evaluation of patients with

esophageal cancer. We found that these indicators have different values in prognosis prediction. The pretreatment level of CAR and mGPS showed an outstanding prediction vale for 5-year OS than other indicators. However, due to the complexity of prognostic prediction, these indicators have satisfactory specificity, but lack high sensitivity. Developing a more robust indicator is meaningful and is the direction of our future research. This meta can provide a basis for our subsequent research.

#Reviewer 3:
Response:
Thank you for your constructive suggestions. On the one hand, there was certain heterogeneity in this study, and we tried to find out the source of heterogeneity as much as possible through subgroup analysis and meta regression to make the results more credible. The results showed that we found most of the sources of heterogeneity, such as sample size, gender distribution, truncation value, etc. However, there are still some sources of heterogeneity that remain unknown. It is reported that some variables such as age, smoking, alcohol drinking, comorbidities, neoadjuvant therapy, and psychological factors, also affect the outcomes, which may cause potential confounding. However, due to the limited variables reported in the paper, we cannot find the source of heterogeneity at present. We added the discussion of these shortcomings and provide possible sources of heterogeneity in Discussion. On the other hand, our study has clinical value and public health significance. With the development of neoadjuvant therapy, patients' prognosis can not be inferred only by postoperative pathological staging in clinical practice, but some easy to obtain and objective auxiliary indicators are more needed. Accumulating literature have shown the prognostic values of inflammation and nutrition-based biomarkers, but with inconsistent findings. Besides, which predictor will perform better is still unknown. Therefore, we conducted this study to evaluate and compare the role of various indicators in prognostic evaluation of patients with esophageal cancer. Our conclusions can guide the clinical use of CAR and mGPS to evaluate the prognosis of patients with esophageal cancer, which have the highest sensitivity and specificity. These indicators can help us identify patients at high risk for survival so that we can regularly follow up and strengthen health management in this population to achieve tertiary prevention.